IDE402.A – Design Graduation Studio VI

Design Milestones - Hot Beverage Preparing Electronic Device

Shared Workspace Filter Coffee Machine

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IDE x AI.26 Common Module

Key Design Milestones

Timeline & Process Visualization

Research

on the environment of shared workspaces and user behavior with AI, observation and surveys.

Problem Framing

Reframed the design challenge from multiple perspectives. Evaluated speed, safety, and shared-space conflicts.

Concept Development

Generated several design alternatives. Explored brewing mechanisms and dispensing systems. Focused on safety and fairness in shared use

Design Process

Design ideation through sketching, doing a user journey map and preparing a mood board to visualize.

Iteration with AI

Used AI to clarify insights, tested concept prompts through Vizcom. Evaluated feasibility and refined structure

Final Direction

Batch brewing system with controlled dispensing, No glass carafe, Visible coffee reservoir

Research Process

Keywords & Summary

- Safety Risks
- Shared Countertop Space
- Shared Worspaces
- Peak-hour Congestion
- Human Density
- Multitasking

This process investigates how coffee is prepared and consumed in shared office and co-working environments, **focusing on the challenges that occur during peak-hour usage.** The research included **a survey with office workers, field observations in co-working spaces, and behavioral analysis of the coffee preparation process.**

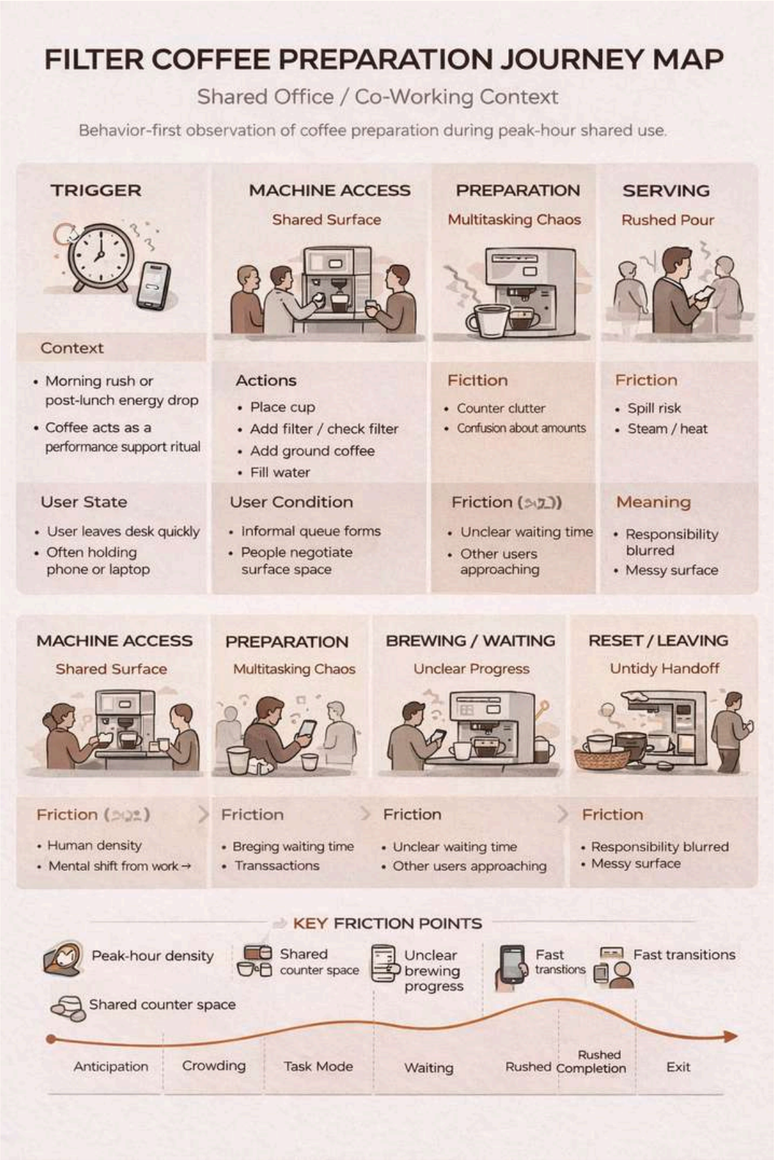
Research Process

Keywords & Summary

Observations revealed that the coffee machine rarely exists as a standalone device. In most offices, **it shares the same countertop with several other appliances.**

Observations revealed that users often operate the machine while **multitasking, such as checking their phones or talking to colleagues, which leads to low-attention interaction.**

Although major safety incidents are rare, **minor spills and rushed movements can occur during busy moments.** Noise and temperature were not identified as major issues.



Research Process

Design Iterations with AI

Role of AI

Selected Impactful Prompts & AI Role Tags

Prompt 1 - Empathizing
Act as role 1: Clarifier. I am designing an electric water heater for offices and co-working spaces.
Can you ask me 10 specific questions to define the environment offices and co-working places have, including: potential risks, safety concerns, challenges and daily chaos.
Please end by listing top 5 aspects that are unknown, that I must observe and ask users.

Prompt 2 - Problem Reframing
Act as Role 2: Reframer. Using these research findings: Coffee is less about identity and more about performance and daily structure, Peak-hour congestion is real, Interaction with the machine happens under distraction and low attention, Cleaning responsibility is usually assigned, Energy awareness is low, Although safety risk is low, it's not impossible, The countertop is usually shared with other machines, Noise is not a primary problem, Temperature satisfaction is moderate. , generate 3 alternative problem framings: speed-first, safety-first and shared-space conflict-first. For each, provide one key risk.

Role of AI

Selected Impactful Prompts & AI Role Tags

Prompt 3 - Problem Statement Critique

Act as Role 4: Critic. Audit this problem statement: During peak hours, coffee machines in shared workspaces are used by numerous distracted users competing for the limited countertop space, which causes congestion and interpersonal issues around the common surface. Identify vague terms, hidden bias, overscope, and claims not supported by empathy. Rewrite a tighter version and explain why it is stronger.

Prompt 4 - Concept Generation

Redesign this filter coffee machine with an internal brewing system, a visible coffee reservoir, and an opaque triangular outlet that dispenses equal portions directly into mugs. (The actual prompt is very long, this is the summarized version.)

Attachment:

Role of AI

Response Summaries

Throughout my project, I used AI as:

1.Clarifier
AI helped interpret research insights and organize them into meaningful themes related to workspace behavior and coffee usage. It broadened my perspective.

2.Re-framer
AI proposed three alternative problem framings focusing on speed, safety, and shared-space conflict.

3.Expander
AI generated design suggestions and functional mechanisms such as hidden brewing systems and controlled dispensing. It helped me find an efficient solution for the top tensions.

4.Critique
AI evaluated the clarity of the problem statement and suggested a tighter, more precise formulation. It made me question my decisions.

Reflections & Decisions

about the filter coffee machine

Throughout the project, I used ChatGPT v4.1 Search Mode and Vizcom.

I mainly used ChatGPT for researching, creating prompts for Vizcom and summarizing my insights that I got from the survey I conducted. I used Vizcom to generate images and to make iterations on my designs.

They both helped accelerate idea exploration and allowed quick visualization of design concepts.

However, many AI-generated images lacked realistic engineering logic, which required critical evaluation and further design refinement.

Through the process, AI functioned more as a collaborative thinking tool rather than a final solution generator.

Final Direction

Accepted, Rejected & Transformed

Accepted

- Batch brewing system
- Visible coffee reservoir
- Mug-based dispensing
- Removable tanks for cleaning

Rejected

- Traditional glass carafe
- Open visible brewing funnel
- Complex user interaction during peak hours

Transformed

- The triangular brewing funnel became a dispensing outlet
- Brewing process moved inside the machine
- Coffee preparation shifted from individual brewing to batch brewing with controlled portions

A shared workspace coffee machine that brews coffee in batches and dispenses equal portions safely into individual mugs, reducing congestion, spills, and interaction complexity.